

Harsha M V

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EDUCATION

George Mason University, Fairfax, VA

Master of Science, Computer Science

Aug 2022 - May 2024

GPA: 3.92 / 4.0

Visveswaraya Technological University, Bengaluru, India

Bachelor of Engineering, Computer Science

Aug 2016 - Jul 2020

GPA: 3.52 / 4.0

CERTIFICATIONS

- Certified Kubernetes Application Developer (CKAD)
- Microsoft Certified: Azure Administrator Associate (AZ-104)
- HashiCorp Certified: Terraform Associate (003)

TECHNICAL SKILLS

Programming Languages: Python, Go, JavaScript, Java, TypeScript, Bash, HCL

Frameworks & Web: FastAPI, NestJS, Spring Boot, Flask, React, NextJs, Vue.js, Node.js, REST APIs

Databases & Storage: PostgreSQL, MySQL, MongoDB, DynamoDB, Cosmos DB, Redis, Elasticsearch, Azure NetApp Files, S3

Cloud Platforms: AWS (EKS, ECS, Lambda, VPC, CloudFormation, IAM, CodePipeline), Azure (AKS, Azure Functions, VNets, Azure OpenAI, API Management, Azure AI Search), Google Cloud (GKE, Vertex AI, Pub/Sub)

Networking: VPC Peering, Transit Gateway, VPN, ExpressRoute, Private Link/Endpoints, NAT Gateway, Azure Firewall, NSG, CIDR Management, DNS, Load Balancers (ALB/NLB)

Generative AI: LLMs (GPT, Gemini, Claude), RAG Pipelines, Embeddings, Vector Search, Agentic AI, Multi-Agent Orchestration, Function Calling, Prompt Engineering, Token Optimization, Guardrails & Safety

Observability & Reliability: Prometheus, Grafana, KEDA, CloudWatch, Azure Monitor, Auto-scaling, High Availability, Disaster Recovery

DevOps Tools: Terraform, Docker, Kubernetes, Helm, Istio, Ansible, Jenkins, GitHub Actions, Argo CD, Packer

EXPERIENCE

Strategy

Senior Software Engineer

Tysons, VA

Jan 2026 – Present

- Built a **Kubernetes Operator** (Java/Spring Boot) with a **15-state finite state machine** for environment lifecycle - provisioning, health monitoring, backup/restore, credential rotation, and rolling restarts via Helm, replacing manual runbooks with single-command automation
- Engineered secure deployments for regulated industries **air-gapped environments** with private registry mirroring (Harbor, Artifactory), **proxy-routed clusters** for controlled external connectivity, and hardened K8s security via **OPA Gatekeeper** policies, PodSecurity standards, and least-privilege RBAC restrictions
- Designed an **event-driven telemetry pipeline** ingesting daily uploads from **100+ customer K8s clusters** - SAS-authenticated Azure Blob uploads triggering EventGrid → Functions → Data Factory pipelines, processing 500K+ records/month into PostgreSQL for operational analytics
- Served as **primary technical point of contact** for 10-15 enterprise customers simultaneously, owning full deployment lifecycle from infrastructure assessment through production hardening across AWS, Azure, GCP, OpenShift, and bare-metal
- Designed and built an **AI-powered RAG assistant** (FastAPI, OpenAI, Azure AI Search) indexing **36k+ pages** across 6 repositories - reducing Tier-1 support tickets by 30-40% and average resolution time by 50%
- Leveraged **Claude Code** as an AI pair-programming agent for architectural design, multi-file refactors, debugging, and test authoring - delivering the AI assistant (design → production) in **4 weeks vs 12 weeks estimated**, a **3x velocity improvement**

Software Engineer

Oct 2024 – Jan 2026

- Architected a multi-cloud BI SaaS platform across AWS, Azure, and GCP using **Terraform, Helm, and Kubernetes**, serving 800+ enterprise tenants and 10,000+ users, contributing to \$150M+ in annual revenue
- Led implementation of a multi-tenant, multi-region **hub-and-spoke network architecture** with VPCs/VNets, Transit Gateways, peering, NAT gateways, firewalls, and private endpoints - enabling secure cross-cloud connectivity with strict tenant isolation

- Developed a scalable multi-LLM AI platform handling 50M+ TPM and 500K+ concurrent API requests, integrating **Azure OpenAI and GCP Vertex AI (Gemini, Llama, Claude)** with regional **APIM** for low-latency analytics bots
- Delivered **HA/DR** across regions and clouds (AWS, Azure, GCP), defining **RTO/RPO** via deployment + replication benchmarks and enabling automated failover for critical networking and storage paths
- Introduced **adaptive endpoint routing** with latency-aware health checks and weighted traffic shifting across LLM endpoints, reducing p95 latency by 50%+ and increasing adoption by 37% while avoiding a 5× cost increase
- Developed a **cross-platform CLI** (PyInstaller) compiled to native binaries for **3 operating systems** with **15+ command groups** - enabling 100+ customers to authenticate (B2C OAuth), deploy/upgrade environments, and run pre-flight validation - replacing fragmented shell scripts and reducing misconfigurations

Associate Software Engineer

Jun 2024 – Oct 2024

- Implemented Halyard, a cloud-agnostic provisioning service (**Java Spring Boot, Redis, GCP Cloud SQL**) automating multi-tenant Kubernetes cluster deployment (GKE, EKS, AKS) for 500+ enterprise customers
- Engineered GitOps-based CI/CD automation using **AWS CodePipeline, Step Functions, and GitHub Actions** with multi-stage Docker builds - reducing build/push time from 45 to 17 minutes across 10+ repos with 98.7% pipelines success rate on 500+ daily executions
- Implemented cloud cost governance across 800+ accounts using **Lambda, API Gateway** with serverless cleanup pipelines, saving \$1M+ annually through resource reclamation with approval workflows and alerting
- Enabled autoscaling using **Prometheus and KEDA** for Kubernetes services with **Grafana** dashboards for CPU/memory visibility, improving resource utilization and reducing peak-time latency by 35%
- Led migration from **Azure NFS File Shares to NetApp Files** for AKS persistent storage, resolving critical DR gaps with zero data loss, enabling snapshot-based cross-region replication and boosting I/O throughput by 25%
- Delivered an **Agentic AI** bot integrated with Microsoft Teams that automated pipeline operations with approval workflows and production guardrails - reducing manual workload by 40% and cutting pipeline errors by 80%

Accenture

Bengaluru, India

Software Engineer

Aug 2020 – Jul 2022

- Containerized and deployed full-stack services with **Docker on Kubernetes**, enabling autoscaling, resilient rollouts, and zero-downtime deployments across multiple AWS-hosted clusters
- Optimized CI/CD pipelines with **Jenkins and Apache Maven**, eliminating 80% of manual build effort and accelerating release velocity across the team
- Authored 50+ **Ansible roles and playbooks** for standardized deployment; refactored 80% of existing roles to enforce compliance with updated organizational standards
- Provisioned and managed **AWS infrastructure (EC2, EBS, security groups, Auto Scaling, Route 53, SNS)** aligned to architecture standards, supporting reliable multi-environment deployments
- Standardized AMI builds via **Packer** integrated into CI pipelines with **Nessus** vulnerability scanning, ensuring hardened base images across all deployments